Real-World Applications

De-Banking, Payments, Tokenization, and Security

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Algorithm Basics and BW

What We'll Cover Today

1. De-banking: What it is and why it matters
2. Cross-border payments: Crypto vs traditional
3. Tokenization of physical assets
4. Fintech infrastructure moving on-chain
5. Security challenges: Hacks and quantum computing
6. Hands-on: Create 🦊 MetaMask wallet + send tokens

Quick Recap: Major Cryptocurrencies

Last class we explored:

- Bitcoin - Digital gold, store of value
- Ethereum - Smart contract platform
- ISO 20022 tokens - Banking compliant (XRP, XLM)
- DeFi tokens - Decentralized finance (UNI, AAVE)
- DePIN tokens - Infrastructure (ICP, XYO)

Today: Real-world applications and hands-on practice

What is De-Banking?

Definition:

- 🏛 When banks refuse service to individuals or businesses
- 🚫 Account closures without clear explanation
- 🔒 Frozen funds and restricted access
- 🔨 Often based on perceived risk or political reasons

Who gets de-banked?

1. Cryptocurrency businesses
2. Cannabis industry (legal in some states)
3. Adult entertainment industry
4. Political dissidents and activists

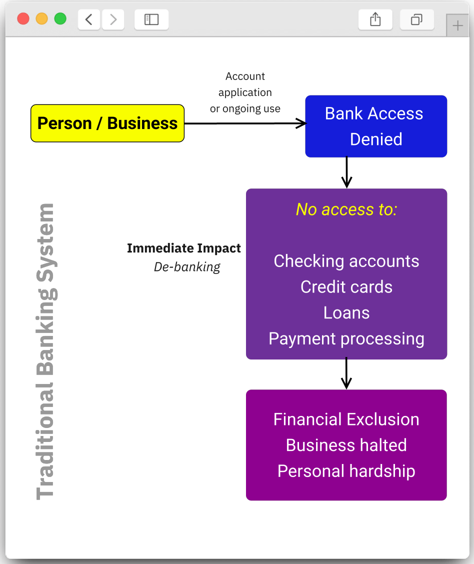
De-Banking Problem Visualized

Why banks de-bank:

- Regulatory compliance burden
- Risk aversion
- $ Not profitable enough
- Political pressure

Impact:

- Can't receive payments
- Can't pay bills
- Business operations halt
- Personal financial exclusion



Real-World De-Banking Examples

1. **Crypto exchanges:** Coinbase, Kraken faced closures
2. **Cannabis businesses:** Legal in state, can't use banks
3. **Activists:** Canadian truckers (2022), Nigeria protests
4. **High-risk merchants:** Payment processors shut down
5. **Individuals:** Wrong political views or associations

Key problem:

- Banks act as gatekeepers to financial system
- Can deny service arbitrarily
- No alternative for those excluded

How Cryptocurrency Solves De-Banking

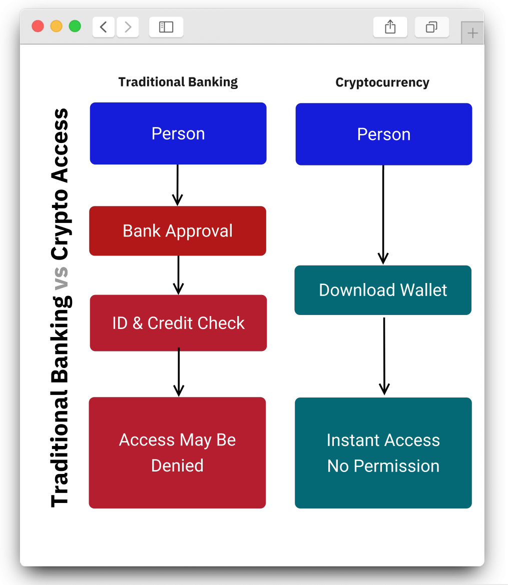
Crypto as alternative:

- **Permissionless:** No approval needed to use
- **Decentralized:** No single entity can block you
- **Borderless:** Works anywhere with internet
- **Censorship resistant:** Transactions can't be stopped

Real solution:

- Cannabis businesses accept Bitcoin
- Activists receive donations via crypto
- De-banked individuals maintain financial access
- Not dependent on bank approval

Traditional Banking vs Crypto Access



Traditional banking:

- ID required
- Credit check
- Bank approval
- Can be denied

Cryptocurrency:

- Download wallet
- Instant access
- No approval needed
- Available to everyone

Cross-Border Payments: The Problem

Sending money internationally is:

- **Slow:** 2-5 business days typical
- **Expensive:** 3-7% fees + FX markup
- **Complex:** Multiple intermediaries
- **Opaque:** Hidden fees, unclear exchange rates
- **Restricted:** Capital controls in some countries

Traditional system:

- SWIFT network - Created in 1973
- Correspondent banking - Multiple hops
- Each bank takes a cut

How SWIFT Works (Traditional)

Sending $1000 from USA to Philippines:

1. Your bank (USA) → SWIFT message
2. Correspondent Bank 1 (intermediary)
3. Correspondent Bank 2 (intermediary)
4. Final bank (Philippines)
5. Recipient receives money

Costs:

- Your bank fee: $25-45
- Intermediary fees: $10-20 each
- Exchange rate markup: 2-4%
- Total cost: $50-80 + poor exchange rate

SWIFT vs Crypto Comparison

SWIFT (Traditional):

- 2-5 days
- $50+ fees
- 4-5 intermediaries
- Business hours only
- ID verification required

Crypto (XRP/USDC):

- 3-60 minutes
- $0.50-5 fees
- Peer-to-peer (no intermediaries)
- 24/7/365
- No permission needed

Crypto is faster, cheaper, and always available

Crypto Cross-Border Payment Flow

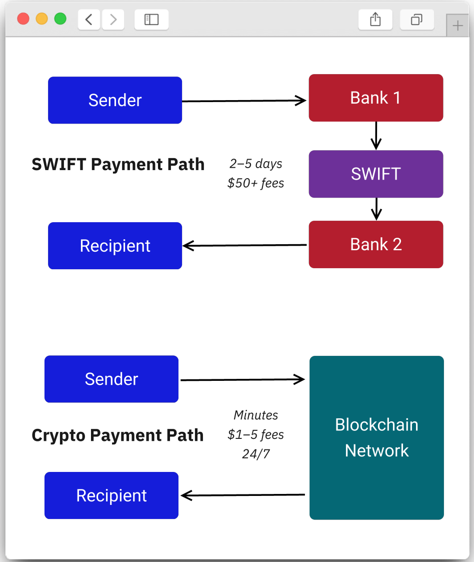
Sending $1000 using USDC stablecoin:

1. Convert USD to USDC ($1-2 fee)
2. Send USDC on blockchain ($0.50-3 fee)
3. Transaction settles in minutes
4. Recipient converts USDC to local currency
5. Total time: 10-30 minutes

Total cost:

- Network fee: $0.50-3
- Exchange fee: 0.5-1%
- Total: $5-15 (vs $50-80 traditional)
- Savings: 70-80%

Payment Flow Comparison



Why crypto wins:

- ⇄ Direct transfer
- No intermediaries
- 24/7 settlement
- $ Lower fees
- True borderless

Use case:

- $600B in remittances sent globally yearly
- Crypto can save billions in fees

Tokenization: Bringing Assets On-Chain

What is tokenization?

- Representing real-world assets as digital tokens
- ⛓ Token on blockchain = ownership proof
- Can be traded, divided, transferred digitally
- Accessible to anyone, anywhere

What can be tokenized?

1. Real estate (buildings, land)
2. Art and collectibles
3. Stocks and securities
4. Precious metals (gold, silver)
5. Invoices and debt

Why Tokenize Physical Assets?

1. **Fractional ownership:** Own part of expensive asset
2. **Liquidity:** Trade 24/7, not just business hours
3. **Global access:** Anyone can invest, not just locals
4. **Transparency:** Ownership recorded on blockchain
5. **Faster settlement:** Minutes instead of weeks

Example:

- \$1M building $\rightarrow$ 1000 tokens at \$1000 each
- Buy 1 token $=$ Own 0.1% of building
- Receive proportional rent payments

Real Estate Tokenization Example

Traditional real estate:

- Need \$100K-\$1M to invest
- Complex paperwork
- Weeks to close
- Limited to local market
- Hard to sell quickly

Tokenized real estate:

- Invest \$100-1000
- Smart contract handles it
- Minutes to buy
- Global investment
- Trade anytime

Democratizes access to real estate investment

Tokenization Benefits and Challenges

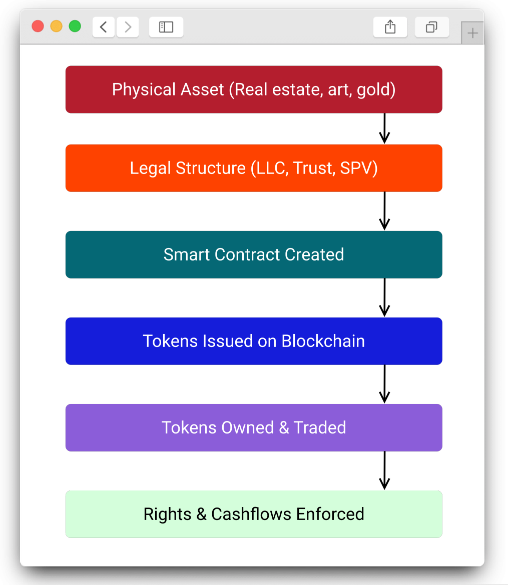
Benefits:

- Democratization - Anyone can invest small amounts
- Increased liquidity for traditionally illiquid assets
- Global market access
- Transparent ownership records

Challenges:

- Legal/regulatory framework unclear
- Who enforces real-world rights?
- Asset custody and verification
- Education needed for adoption

Tokenization Process



Steps to tokenize:

1. Identify asset
2. Legal structure (LLC, trust)
3. Create smart contract
4. Issue tokens on blockchain
5. Market and sell tokens
6. Manage ongoing operations

Real projects:

- RealT - Tokenized US properties
- Paxos Gold - Gold-backed tokens

Fintech Infrastructure Moving On-Chain

What does this mean?

- Financial services built on blockchain
- Infrastructure on decentralized networks
- Smart contracts replacing traditional systems
- DeFi protocols as new financial rails

Examples of fintech on-chain:

1. Lending/borrowing (Aave, Compound)
2. Trading (Uniswap, dYdX)
3. Payments (Lightning Network, Polygon)
4. Savings (Anchor, Yearn)

Why Move Fintech On-Chain?

1. **Always available:** 24/7/365 no downtime
2. **Composability:** Build on top of other protocols
3. **Transparency:** All transactions auditable
4. **Global:** Anyone can access from anywhere
5. **Automation:** Smart contracts execute automatically

Example:

- Traditional loan: Apply, wait days, bank approval
- DeFi loan: Deposit collateral, instant approval, borrow immediately

Traditional vs On-Chain Fintech

Traditional fintech:

- Centralized servers
- Business hours
- Heavy regulation
- KYC/AML required
- Controlled by company

On-chain fintech:

- Decentralized blockchain
- 24/7 availability
- Code is the regulation
- Permissionless access
- Controlled by community

Both have advantages - hybrid future likely

Security Challenges: Hacks and Exploits

Blockchain is secure, but vulnerabilities exist:

1. **Smart contract bugs:** Code vulnerabilities exploited
2. **Bridge hacks:** Cross-chain bridges attacked
3. **Private key theft:** Phishing, malware
4. **Exchange hacks:** Centralized exchanges compromised
5. **Social engineering:** Users tricked into giving access

Notable hacks:

- Ronin Bridge: $625M (2022)
- Poly Network: $611M (2021, returned)
- FTX collapse: $8B+ (2022)

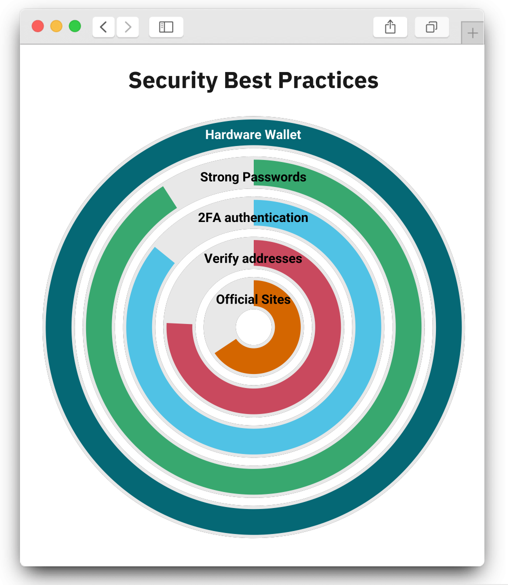
Common Attack Vectors

- **</> Smart contract exploits:** Reentrancy, overflow bugs
- **Phishing:** Fake websites steal private keys
- **Malware:** Keyloggers, clipboard hijackers
- **Rug pulls:** Developers abandon project with funds
- **Social engineering:** Impersonation, fake support

Protection:

- Use hardware wallets for large amounts
- Verify contract addresses carefully
- Never share private keys
- Use reputable, audited protocols

Security Best Practices



Essential practices:

1. Hardware wallet for large amounts
2. Strong, unique passwords
3. Enable 2FA everywhere
4. Verify all addresses
5. Bookmark official sites
6. Regular security audits

Quantum Computing Threat

What is quantum computing?

- Computers using quantum mechanics
- Exponentially faster for certain problems
- Could break current encryption methods
- Still years away from practical threat

Threat to blockchain:

- Could crack private keys (theoretically)
- Mining algorithms could be broken
- Signature schemes vulnerable

Quantum Resistance Solutions

Blockchain community is preparing:

1. **Quantum-resistant algorithms:** New cryptography
2. **Protocol upgrades:** Planned transitions
3. **Timeline:** 10-20 years before real threat
4. **Research:** Active development of solutions

Projects working on this:

- Algorand - Quantum-resistant features
- IOTA - Quantum-ready architecture
- Research ongoing in Bitcoin/Ethereum communities

Key point: Not an immediate threat, solutions being developed

Hands-On Activity: Create MetaMask Wallet

Now let's get practical!

What is MetaMask?

- Browser extension crypto wallet
- Works with Ethereum and compatible chains
- Most popular wallet (30M+ users)
- Free to download and use

Hands-On Activity: Create MetaMask Wallet

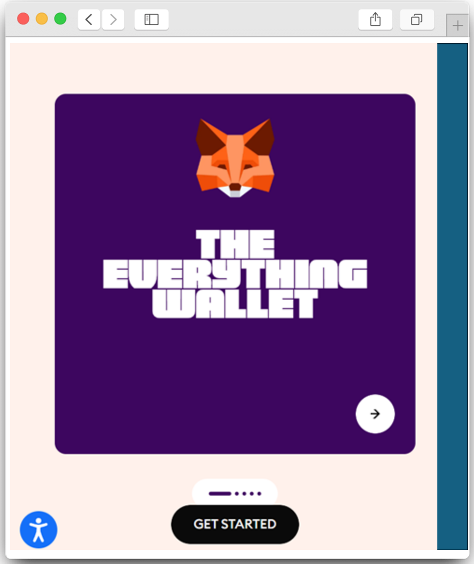
What we'll do:

1. Install MetaMask browser extension
2. Create new wallet
3. Secure your seed phrase
4. Receive test tokens (from instructor)
5. Send tokens to classmate
6. View transaction on blockchain explorer

Step 1: Install MetaMask

Follow along:

1. Open Chrome/Brave/Edge browser
2. Go to `https://metamask.io`
3. Click "Download"
4. Select "Chrome" (or your browser)
5. Click "Add to Chrome"
6. Click "Add extension"
7. MetaMask fox icon appears



Step 2: Create New Wallet

Click the MetaMask extension icon:

1. Click "Create a new wallet"
2. Agree to terms
3. Create strong password (12+ characters)
4. Confirm password
5. Watch security video (optional but recommended)
6. → Click "Next"

Important:

- Password is local - only for this device
- Seed phrase is the real backup

Step 3: Secure Your Seed Phrase

MOST IMPORTANT STEP:

1. Click “Reveal Secret Recovery Phrase”
2. Write down all 12 words IN ORDER
3. Double-check spelling
4. Store in safe place (not digitally!)
5. Click “Next”
6. Enter words to confirm

CRITICAL WARNING:

- Anyone with these 12 words controls your wallet
- Never share with anyone
- Never store digitally (no photos, no cloud)
- Write on paper, store in safe place

Understanding Your Wallet Address

You now have:

- A wallet address (like: `0x742d35Cc...`)
- Private key (derived from seed phrase)
- Public address (safe to share)

Copy your address:

1. Click account name at top
2. Click copy icon
3. Address copied to clipboard
4. Paste in chat/email to instructor

Sender will send everyone small amount of test tokens

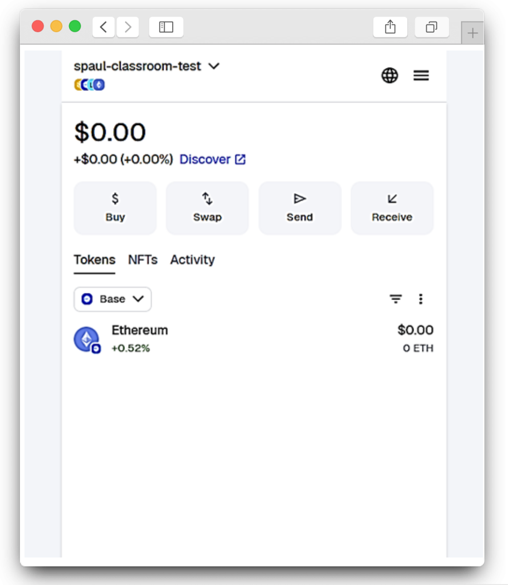
Step 4: Receive Test Tokens

Waiting for tokens:

1. Sender sends to your address
2. Wait 30-60 seconds
3. Click refresh if needed
4. Tokens appear in wallet!

What you'll receive:

- Small amount of test tokens
- Enough to make a transaction
- Not real money (testnet or dust amount)



Step 5: Send Tokens to Classmate

Now let's make a transaction:

1. Partner with a classmate
2. Exchange wallet addresses
3. Click "Send" in MetaMask
4. Paste their address
5. Enter small amount (half your balance)
6. Verify address is correct
7. Click "Next" then "Confirm"
8. Wait 30-60 seconds for confirmation

You just made your first crypto transaction!

Step 6: View on Blockchain Explorer

See your transaction on the blockchain:

1. Click on transaction in MetaMask
2. Click "View on block explorer"
3. Opens Etherscan (or similar)
4. See full transaction details
5. Transaction hash (unique ID)
6. Confirmation status

What you can see:

- From/To addresses
- Amount sent
- Fee paid
- Timestamp
- All public and permanent!

What You Just Experienced

1. Created a self-custody wallet
2. Secured it with seed phrase
3. Received cryptocurrency
4. Sent crypto to another person
5. Verified on blockchain explorer

Key learnings:

- You control your wallet (not a bank)
- Transactions are fast and direct
- Everything is transparent and auditable
- Seed phrase is crucial - guard it carefully

Key Takeaways

1. De-banking is real - crypto provides alternative access
2. Cross-border payments: Crypto is faster, cheaper than SWIFT
3. Tokenization democratizes access to physical assets
4. Fintech moving on-chain: 24/7, transparent, global
5. Security matters: Use best practices, quantum threat distant
6. Self-custody = You control your money

Remember:

- With great power comes great responsibility
- Your keys, your crypto - protect them!

Questions and Discussion